

What `sync()` Certifies: A Consumer-Relative, Witnessed Reading of ZooKeeper Staleness

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Abstract—A ZooKeeper client may read from any server, and a follower answers from its own applied state—so a local read can be stale. ZooKeeper offers `sync()` to force a read to reflect the latest committed state. We use this familiar mechanism as a compact introduction to *Observation Theory* (OT): the thesis that reliability is a property of the relation between a system’s error and *what a particular consumer reads*, and that any honest freshness claim needs an independent *witness*. In OT terms a local read is a *certificate* (this follower is fresh enough), the ZooKeeper transaction id (`zxid`) is its *witness*, and `sync()` is a witnessed correction. We measure, on a three-node ensemble under a pre-registration whose pass/fail bars are fixed before the runs, the *false-clear rate* of the local-read certificate. The result is a clean statement of consumer-relativity: on the *same* lagging follower, a hot-footprint reader false-clears $\approx 99\%$ of the time while a cold-footprint reader reads $\approx 1\%$ stale; graded against the `zxid`, `sync()` drives the false-clear to zero. Staleness is thus a property of the reader’s footprint, not of the replica—and `sync()` certifies not “the follower is caught up” but “the consumer’s footprint is caught up.”

Index Terms—ZooKeeper, staleness, read consistency, `sync`, observation theory, pre-registration.

I. INTRODUCTION

ZooKeeper [1] serves reads locally: a client connected to a follower is answered from that follower’s applied state. Writes are linearizable through the leader via Zab [2], but a local read is not—it reflects whatever the follower has applied, which may trail the leader. For readers that need the latest state, ZooKeeper provides `sync()`, which flushes the leader-to-follower channel so a subsequent read reflects at least the committed state at the time of the `sync`.

This paper takes that familiar mechanism as the smallest possible worked example of a general idea we call *Observation Theory* (OT). The question OT asks of any freshness monitor is: *when it says a read is “fresh enough,” how often is it wrong, and for whom?* We show that for ZooKeeper the answer is sharply consumer-relative—the same follower is 99% wrong for one reader and 1% wrong for another at the same instant—and that the fix is not a better lag bound but a witness. Our contributions are (i) a one-section, self-contained introduction to OT (Section II); (ii) its instantiation for ZooKeeper reads (Section III); and (iii) a sealed, reproducible measurement of the local-read false-clear rate (Sections IV–V).

II. OBSERVATION THEORY IN ONE SECTION

OT begins from the observation that reliability is not a property of an object alone but of a *relation* among the

object, an observer, and a resolution budget. An *observer*, or *consumer*, is a map $\mathcal{C}$ from a representation to a task output, together with a metric on that output. If the representation carries an error δ with second moment $\Sigma = \mathbb{E}[\delta\delta^\top]$, a second-order expansion of the consumer’s felt distortion isolates a single controlling quantity,

$$d_{\mathcal{C}} = \text{tr}(P_{\mathcal{C}} \Sigma), \quad (1)$$

where $P_{\mathcal{C}} \succeq 0$ is the *read operator*—the pullback of the output metric through $\mathcal{C}$, encoding *which directions of the error the consumer can actually feel*. Ordinary reconstruction error is the isotropic corner $P_{\mathcal{C}} = I$; a real consumer reads a subspace, and (1) weights the error by it. The same quantity governs consumer-relative lossy compression [3] and, as we now show, freshness.

Two consequences frame everything below. First, a *certificate*—a claim that a representation is good enough for the consumer—is only meaningful relative to $P_{\mathcal{C}}$; a scalar bound that ignores $P_{\mathcal{C}}$ serves no one in general. Second, a certificate should be *witnessed*: graded against an independent measurement of the true error Σ , not inferred from a proxy. A certificate that is both consumer-relative (reads $P_{\mathcal{C}}$) and witnessed (grades against a measurement of Σ) has a well-defined *false-clear rate*—the rate at which it clears a read whose felt error exceeds tolerance. OT’s empirical program measures that rate across systems; this paper measures it for ZooKeeper.

III. ZOOKEEPER READS AS CERTIFICATES

The mapping is direct. The replica drift Σ is the per-znode staleness of a follower: for each znode, how far its applied version trails the leader’s committed version. The consumer’s read operator $P_{\mathcal{C}}$ is its *footprint*—which znodes it reads, and how heavily. The felt staleness $\text{tr}(P_{\mathcal{C}}\Sigma)$ is then simply whether *the znodes this reader touches* are behind.

The witness is already in the protocol. Every znode carries the `zxid` of its last modification (`mzxid`); the leader’s committed `mzxid` is the ground truth, and the follower’s is what a local read returns. A local read is stale exactly when the follower’s `mzxid` for a footprint znode is behind the leader’s—an *observable* condition, not an inference. And `sync()` is the witnessed correction: it forces the follower to the leader’s `zxid` before the read, zeroing the felt staleness for that read.

TABLE I
XPROTO-ZK SHAKEDOWN MEASUREMENT (PRE-SEAL, SEEDS $\{0, 1, 2\}$);
THE SEALED GRADED RUN USES DISJOINT SEEDS AND IS EXPECTED TO
MATCH. “NAIVE-HOT”/ “NAIVE-COLD” ARE THE LOCAL-READ STALE
RATES FOR A HOT/COLD FOOTPRINT ON THE *same* DELAYED FOLLOWER;
“`sync()`” IS THE WITNESSED POLICY.

seed	naive-hot	naive-cold	<code>sync()</code>	zxid gap
0	0.990	0.010	0.000	12.3
1	0.997	0.011	0.000	12.5
2	0.994	0.011	0.000	12.7

The naive certificate—“a local read is fresh enough”—ignores P_C : it clears every local read. OT predicts it will false-clear at a rate set by the reader’s footprint, and that a `zxid`-witnessed policy (of which `sync()` is the deployed form) will not. We test both.

IV. METHOD: A SEALED MEASUREMENT

Substrate. A three-node ZooKeeper 3.9 ensemble (one leader, two followers). To place the system in a staleness regime we delay the Zab commit stream to one follower by 200 ms (a `netem` delay on the leader’s egress toward that follower); the follower’s client port is undelayed, so local reads stay prompt while its applied state lags—the coordination-service analogue of a delayed database replica. A writer updates a hot set of `znodes` every ~ 12 ms and a cold set every ~ 3.6 s.

Policies. Over one run we sample two policies on the delayed follower: **naive** (trust the local read) and **witnessed** (`sync()` the path, then read). For each sampled read we capture the leader’s `mzxid` *before* the follower read, so a write committing during the read is not counted stale (in-flight writes do not disqualify the follower). We report the false-clear (stale) rate for a **hot** footprint and, on the *same* follower, for a **cold** footprint.

Discipline. Bars are committed in a pre-registration before the graded run, with a cooling-off enforced in code and graded seeds disjoint from those used to calibrate. The bars: the naive hot-footprint false-clear ≥ 0.25 ; the witnessed false-clear ≤ 0.10 ; dominance; and manipulation checks that the follower genuinely lags, that the cold footprint on the same follower is fresh (so the follower is sane and the staleness is footprint-relative), and that the run is non-degenerate.

V. RESULTS

Table I reports the pre-seal shakedown; the sealed graded run (on seeds disjoint from these) is expected to match and will replace it. On the same delayed follower, the hot-footprint local read is stale $\approx 99\%$ of the time, while the cold footprint reads $\approx 1\%$ stale; `sync()` holds the false-clear at zero; the mean leader-follower `zxid` gap is ~ 12 . The bars and all manipulation checks pass on every seed.

The headline is the two middle columns. Same replica, same instant, same induced lag: a $99\times$ difference in false-clear, produced by nothing but *what the reader reads*. This is $\text{tr}(P_C\Sigma)$ made concrete: the drift Σ is identical for both readers; the

felt staleness differs because P_C differs. A single “follower lag $< T$ ” monitor cannot serve both—set T to protect the hot reader and the cold reader is needlessly blocked; set it for the cold reader and the hot reader is served stale. The witnessed policy sidesteps the dilemma: it reads P_C (only the footprint) and grades against Σ (the `zxid`). `sync()` certifies not “the follower is caught up” but “the consumer’s footprint is caught up.”

VI. DISCUSSION

What this says about bounded-staleness reads. Systems commonly expose a staleness bound as a scalar (a lag threshold, a bounded-staleness read level). The result argues that the useful bound is per-footprint and witnessed: probabilistically bounded staleness [4] calibrates the distribution but globally; Spanner’s commit-wait [5] and session guarantees [6] are witnessed or consumer-relative but not jointly calibrated per footprint. The `zxid` already provides the witness; the missing piece is to read P_C .

The broader program. The same certificate/witness/false-clear grammar recurs across substrates: WAL- and oplog-witnessed footprint certificates in PostgreSQL and MongoDB, HARQ-witnessed adaptation certificates in 5G/6G radio access, and BMP-witnessed reachability certificates in inter-domain routing [7]. ZooKeeper is the coordination-service case, and—because its witness (`zxid`) and its correction (`sync()`) are already in the protocol—the cleanest. Age of information [8] bounds staleness on the update channel; $\text{tr}(P_C\Sigma)$ measures its consequence for a specific reader, which is what a consistency SLA must bound. In CAP/PACELC terms [9], [10], the footprint certificate quantifies the interior of the else-latency regime: it prices the consistency each reader actually needs and pays latency (a `sync()`) only when the footprint demands it.

VII. LIMITATIONS AND CONCLUSION

The measurement is a link-level ensemble with induced lag rather than a production trace; watch-based reads, larger ensembles, and observer read-scaling are natural extensions, and the footprint here is a set of `znodes` rather than a general read operator. Within those bounds the conclusion is clean and, we hope, useful to ZooKeeper’s own users: a local read’s freshness is a property of the reader’s footprint, not of the follower; the `zxid` is a sufficient witness; and `sync()` is best understood as certifying the consumer’s footprint. More broadly, ZooKeeper is a particularly legible entry point to Observation Theory—a system whose designers already built both the witness and the correction the theory asks for.

REPRODUCIBILITY

The pre-registration, the coded verdict runner (with the cooling-off and substrate guards), the ensemble scripts, and the graded record are archived with the program [11]; the substrate is unmodified ZooKeeper 3.9.

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